SYNOPSIS

for

Identity Security Framework for Healthcare

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Submitted to

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YENEPOYA INSTITUTE OF ARTS, SCIENCE, COMMERCE AND MANAGEMENT
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I. Title of the Project

Identity Security Framework for Healthcare using Identity and Access Management (IAM), Multi-Factor Authentication (MFA), Single Sign-On (SSO), and Blockchain Technology

II. Statement of the Problem

The healthcare sector is undergoing rapid digital transformation, with hospitals and medical institutions increasingly relying on electronic systems to manage patient records, diagnostics, billing, and clinical workflows. While this digitalization improves efficiency and accessibility, it also introduces significant cybersecurity risks. Healthcare systems store highly sensitive and confidential patient information, making them attractive targets for cyberattacks such as unauthorized access, identity theft, ransomware attacks, and data breaches.

Existing authentication mechanisms in many healthcare systems primarily depend on traditional username and password-based access, which are highly vulnerable to attacks such as phishing, credential stuffing, and brute-force attacks. Additionally, the absence of centralized identity management across multiple healthcare applications leads to fragmented access control, poor visibility of user activities, and increased risk of insider threats.

Another major challenge is the lack of secure and tamper-proof audit mechanisms to track user activities. In critical environments like healthcare, it is essential to maintain transparency and accountability regarding who accessed what data and when. However, conventional logging systems can be altered or compromised, leading to a lack of trust and compliance issues.

The problem is further intensified by the use of multiple independent systems such as Electronic Medical Records (EMR), Laboratory Information Systems, and Billing Systems, each requiring separate authentication. This not only reduces user convenience but also increases security risks due to password reuse and poor credential management practices.

The core problem lies in the absence of a comprehensive and secure Identity Security Framework that can provide centralized identity management, strong authentication mechanisms, seamless access across systems, and secure audit trails.

Key Challenges:

1. Managing multiple user identities with role-based access control
2. Implementing strong authentication (MFA)
3. Integrating Single Sign-On (SSO)
4. Ensuring tamper-proof logging using Blockchain

5. Handling scalability and performance
6. Protecting sensitive healthcare data
7. Designing a user-friendly system

III. Why this Particular Topic Chosen?

The choice of this topic is based on the increasing importance of cybersecurity in healthcare systems and the need to protect sensitive patient data.

• **Relevance and Importance:**
Healthcare data security is critical in modern digital systems.

• **Challenging and Engaging:**
The project involves IAM, MFA, SSO, and Blockchain integration.

• **Personal and Professional Growth:**
Enhances skills in cybersecurity, backend development, APIs, and cloud technologies.

• **Business and Community Impact:**
Improves data privacy, reduces cyber risks, and increases trust in healthcare systems.

• **Future Scope:**
Can be extended with biometric authentication.

IV. Objective and Scope

➢ Objectives

- To design a centralized Identity and Access Management (IAM) system
- To implement Multi-Factor Authentication (MFA) using OTP/TOTP
- To enable Single Sign-On (SSO) using OAuth 2.0 and OpenID Connect
- To implement Role-Based and Permission-Based access control (Read/Write)
- To maintain secure and tamper-proof logs using Blockchain
- To enhance system security using adaptive authentication

- **Scope**

  - Applicable for hospitals and healthcare institutions
  - Focuses on authentication, authorization, and auditing
  - Can be integrated with EMR, Lab, and Billing systems
  - Designed as a modular and scalable system
  - Deployable on cloud platforms (AWS/Azure)
  - Limited to prototype-level implementation

V. Methodology

The project follows the Agile Software Development Methodology, which supports iterative and flexible development.

Key Features

- Iterative development using sprints
- Continuous testing and integration
- Regular feedback and improvements
- Flexibility to adapt to changes

Development Phases

1. Requirement Analysis
2. System Design (HLD & LLD)
3. Development
4. Testing
5. Deployment

Justification

Agile is suitable because it allows:

- Continuous improvement
- Early detection of issues
- Better handling of complex security systems

VI. Process Description

The system follows a modular architecture with two main entities: Users and Employees, managed with roles and permissions.

Modules

1. User Management Module

- Handles registration of patients and employees
- Assigns roles (Admin, Doctor, Nurse, Lab Staff)

2. Employee & Permission Module

- Admin creates employees
- Assigns roles and permissions (Read / Write)

3. Authentication Module

- Validates login credentials
- Generates secure JWT tokens

4. MFA Module

- Provides OTP / TOTP verification

5. SSO Module

- Enables single login across multiple systems

6. Access Control Module

- Implements RBAC + Permission-based control

7. Blockchain Logging Module

- Stores tamper-proof activity logs

Permission Types

- Read Permission: Allows employees to view system data without making any changes.
- Write Permission: Allows employees to create, update, or modify system data based on assigned roles.

Process Flow

1. User enters login credentials
2. System verifies credentials
3. MFA verification is performed

4. Token is generated
5. Role and permissions are checked
6. Access is granted
7. Activity is logged in Blockchain

The system design can be further represented using Data Flow Diagrams (DFD) and flowcharts to illustrate the flow of information between modules.

VII. Resources and Limitations

Resources

Hardware

- Computer/Laptop
- Internet connection

Software

- Frontend: React.js
- Backend: Node.js
- Database: PostgreSQL
- Tools: VS Code, Postman
- Blockchain: Ethereum (Ganache)
- Cloud: AWS / Azure

Limitations

- Prototype-level implementation
- Blockchain is simulated
- No biometric authentication
- Depends on internet connectivity
- Limited scalability optimization

Future Improvements

- Biometric authentication
- AI-based security
- Real blockchain deployment
- Improved UI/UX

VIII. Testing Technologies Used

Testing Types

- Unit Testing
- Integration Testing
- System Testing
- UI Testing
- Security Testing

Techniques

- Black Box Testing
- White Box Testing

Tools

- Postman
- Browser Developer Tools

Testing Outcome

- Secure authentication verified
- Proper role access ensured
- System reliability validated

IX. Conclusion

The proposed system provides a secure and efficient Identity Security Framework for healthcare using IAM, MFA, SSO, and Blockchain.

Achievements

- Strong authentication system
- Centralized identity management
- Secure access control
- Tamper-proof logging
- Improved user experience

Innovation

Combination of:

- IAM + MFA + SSO
- RBAC + Permission system
- Blockchain logging

Future Scope

- Biometric authentication
- Full healthcare system integration

This project serves as a strong foundation for developing secure, scalable, and reliable healthcare systems in the future.